wartime physics research program, employing thousands and costing millions) and consulted for the Royal Air Force. Before the war, he had worked at the University of Michigan and had planned to retire from research and devote himself to teaching full-time. After the war, he changed his mind. “I felt caught up in the violent upsurge of everything associated with physics that had followed Hiroshima,” Goudsmit recalled, “and I wanted to be more closely associated with it than seemed possible on a university campus.” Goudsmit became the head of the physics department at Brookhaven National Laboratory, one of the newly minted government-run pure research labs. Yet, despite his administrative post in the new edifice of “Big Science,” Goudsmit remained uneasy about the changes in his field. “The conditions we work under today certainly aren’t hastening [a] breakthrough,” he said in 1953.

> A quarter of a century ago we could exchange ideas in Bohr’s study with no government secrets, weapons programs, or spy cases to bother us.… None of us were distracted by offers to become college presidents or big wheels in industry, and governments didn’t give a hoot about physicists. There was no trying to elbow one’s way to power, for the simple reason that there wasn’t any place to exercise power. No huge laboratories, no military projects.… We all felt that we belonged to a sort of lodge, with a worldwide membership of only four hundred or so, and everyone knew everyone else well—or at least knew what everyone else was doing. Now four times that number will turn up for a meeting of just American physicists, and most of them will be strangers to each other.

Research into the meaning of quantum physics was one of the casualties of the war. With all these new students crowding classrooms around the country, professors found it impossible to teach the philosophical questions at the foundations of quantum physics. Before the war, courses in quantum physics on both sides of the Atlantic, like Heisenberg’s in Leipzig and Oppenheimer’s in Berkeley, spent a great deal of time on conceptual issues. Textbooks and exams from the prewar period asked students to write detailed essays on the nature of the uncertainty principle and the role of the observer in the quantum world. But, with ballooning class sizes, detailed discussion of philosophy became all but impossible. “With these subjects [such as uncertainty, complementarity, and causality] lecturing is of little avail,” complained a physics professor at the University of Pittsburgh in